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CO₂-Surfactant Composite System Screening and Microscopic Enhanced Oil Recovery Mechanism Study

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Carbon Capture, Utilization, and Storage (CCUS) involves separating and capturing CO₂ from industrial emissions or the atmosphere, then transporting it via tankers, pipelines, or ships to sites for use or storage (Jiang et al. 2020, Chen et al. 2022, Yan and Zhang 2019). This technology reduces overall emission reduction costs and provides flexibility for achieving greenhouse gas reduction targets, making it a key technology for reaching carbon peak and carbon neutrality. As a CCUS technique, CO₂-enhanced oil recovery (CO₂-EOR) involves injecting CO₂ into underground oil reservoirs to boost crude oil recovery while enabling large-scale geological storage, and has been widely applied internationally (Yan and Zhang 2019, Yang et al. 2024). With growing global energy demand and depleting traditional oil fields, improving oil recovery has become crucial for the oil and gas industry. CO₂-EOR technology, which uses CO₂ to reduce oil phase viscosity and enhance oil-gas interfacial activity, has been widely applied and studied abroad. The U.S., one of the earliest countries to initiate CO₂-EOR projects, has relatively mature technology and has achieved significant recovery rate improvements (Hill et al. 2020, Webb et al. 2024). CO₂-EOR can be divided into miscible and immiscible flooding. Miscible flooding forms a single mixed phase under reservoir conditions for efficient crude oil displacement, while immiscible flooding enhances recovery rates through mechanisms like viscosity reduction, oil expansion, and light component extraction. Emerging technologies like water-alternating-gas injection are also being explored to further improve CO₂-EOR efficiency. China's research on CO₂-EOR started later but has developed rapidly. Currently, pilot tests and industrial applications of CO₂-EOR have been conducted in various oil fields, such as the Daqing and Shengli oil fields. Studies indicate that the minimum miscibility pressure (MMP) of CO₂ with crude oil is a key factor affecting CO₂-EOR performance (Adel et al. 2018, Chen et al. 2014). By adjusting formation pressure and optimizing CO₂ injection parameters, the MMP can be lowered to enhance recovery efficiency. Researchers are also actively exploring suitable CO₂-EOR solutions for different reservoir types, such as low-permeability and shale oil reservoirs (Luo 2024).

Scholars have conducted early studies on improving recovery rates using CO₂ and surfactants, achieving certain results. For example, adding surfactants can reduce interfacial tension between CO₂ and crude oil, improving compatibility and oil displacement efficiency (Dowlati et al. 2023). Some studies also found

that specific surfactants can synergistically act with CO₂ to enhance its solubility in the oil phase, further boosting crude oil recovery. China has also made progress in this field. Researchers have investigated the compatibility of various surfactants with CO₂, screened out suitable CO₂-surfactant composite systems for different reservoir conditions, and verified their effectiveness in improving recovery rates through laboratory simulations and field tests. Common laboratory methods for measuring CO₂-crude oil miscibility pressure include the capillary tube method, bubble characteristics analysis (BRA), and interfacial tension disappearance method (VIT) (Zhang et al. 2015, Zhang et al. 2020, Safaei et al. 2023). The capillary tube experiment, a widely recognized and accepted method for determining minimum miscibility pressure (MMP), involves conducting displacement simulations in a long, thin tube model. In China, the MMP assessment standard is based on injecting gas at 1.2 times the pore volume in the capillary model (Yu et al. 2024). When the recovery rate reaches 90% - 95%, the system is considered miscible. By plotting the relationship between recovery rate and displacement pressure for the immiscible and miscible phases, the curve's inflection point is identified as the MMP. If higher-pressure experiments show no significant increase in recovery rate compared to the MMP, the system is deemed miscible. To address the time-consuming and costly nature of the capillary tube experiments, some scholars have proposed the bubble characteristics analysis method, which determines MMP by measuring the rise time and velocity of bubbles in gas-liquid mixtures under different pressure conditions (Yuan et al. 2004, Song et al. 2024). Compared to the traditional capillary tube method for measuring MMP, the interfacial tension disappearance method offers the advantages of faster experimentation and lower costs.

However, it is more subjective and requires highly precise instruments. In recent years, researchers have explored surfactants soluble in CO₂ and studied their impact on CO₂-crude oil miscibility pressure. This study selected four surfactants: NP9 (nonylphenol ethoxylate), DY604 (polyoxyethylene acetylene glycol), SPO5 (polyoxyethylene-polyoxypropylene copolymer), and TXIB (cyclohexyl-2-ethylhexanoate) (Gong et al. 2024, Kuang et al. 2021, Zhang, Gong, et al. 2024, Zhang et al. 2020). All these surfactants have unique molecular structures that enable interaction with CO₂ and crude oil, reduce interfacial tension, and enhance oil displacement efficiency in reservoirs. DY604's acetylene group and polyoxyethylene chain form an amphiphilic structure. The acetylene group's high electron-density region can interact with CO₂'s electron-deficient carbon atoms via dipole-quadrupole interactions, significantly increasing CO₂ solubility. The polyoxyethylene chain also enhances affinity with crude oil components through hydrogen bonding. NP9's nonylbenzene structure can efficiently embed into the asphaltene/resin layers of crude oil, disrupting heavy component aggregation through π - π stacking and enhancing CO₂'s asphaltene extraction ability. In SPO5, the polyoxypropylene (PO) segment becomes more hydrophobic at high temperatures, while the polyoxyethylene (EO) segment remains hydrophilic. This thermally sensitive balance maintains a low micelle aggregation number in reservoirs at 80–120°C, preventing high-temperature precipitation. TXIB, as an ester small molecule, can insert into the surfactant micelle's fence layer, expanding the micelle core volume and increasing the solubilization of light hydrocarbons (C₆–C₁₀). This study represents a significant exploration in this field (Sun et al. 2019). By selecting various gas-soluble surfactants, the research investigated their solubility in CO₂ and their impact on CO₂-crude oil miscibility pressure, further deepening the understanding of the mechanisms by which CO₂ and surfactants enhance oil recovery (Liang et al. 2023).

This study used microscopic visualization experiments to measure the miscibility pressure of CO₂-surfactant composite systems with crude oil. Similar to the interfacial tension disappearance method but with differences, this approach allows intuitive observation of changes in oil-gas interface phenomena as pressure increases. Pressure control can be precise to three decimal places. When the pressure increases to a certain level and the oil-gas interface disappears, the pressure at this point is considered the minimum miscibility pressure of the CO₂-surfactant composite system with crude oil. The study also investigated crude oil

recovery under high-temperature and high-pressure conditions using the composite system, clarifying its mechanisms for improving recovery rates.

Introduction

Carbon Capture, Utilization, and Storage (CCUS) involves separating and capturing CO₂ from industrial emissions or the atmosphere, then transporting it via tankers, pipelines, or ships to sites for use or storage (Jiang et al. 2020, Chen et al. 2022, Yan and Zhang 2019). This technology reduces overall emission reduction costs and provides flexibility for achieving greenhouse gas reduction targets, making it a key technology for reaching carbon peak and carbon neutrality. As a CCUS technique, CO₂-enhanced oil recovery (CO₂-EOR) involves injecting CO₂ into underground oil reservoirs to boost crude oil recovery while enabling large-scale geological storage, and has been widely applied internationally (Yan and Zhang 2019, Yang et al. 2024). With growing global energy demand and depleting traditional oil fields, improving oil recovery has become crucial for the oil and gas industry. CO₂-EOR technology, which uses CO₂ to reduce oil phase viscosity and enhance oil-gas interfacial activity, has been widely applied and studied abroad. The U.S., one of the earliest countries to initiate CO₂-EOR projects, has relatively mature technology and has achieved significant recovery rate improvements (Hill et al. 2020, Webb et al. 2024). CO₂-EOR can be divided into miscible and immiscible flooding. Miscible flooding forms a single mixed phase under reservoir conditions for efficient crude oil displacement, while immiscible flooding enhances recovery rates through mechanisms like viscosity reduction, oil expansion, and light component extraction. Emerging technologies like water-alternating-gas injection are also being explored to further improve CO₂-EOR efficiency. China's research on CO₂-EOR started later but has developed rapidly. Currently, pilot tests and industrial applications of CO₂-EOR have been conducted in various oil fields, such as the Daqing and Shengli oil fields. Studies indicate that the minimum miscibility pressure (MMP) of CO₂ with crude oil is a key factor affecting CO₂-EOR performance (Adel et al. 2018, Chen et al. 2014). By adjusting formation pressure and optimizing CO₂ injection parameters, the MMP can be lowered to enhance recovery efficiency. Researchers are also actively exploring suitable CO₂-EOR solutions for different reservoir types, such as low-permeability and shale oil reservoirs (Luo 2024).

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Methodology

Experimental Materials

This study selected four types of surfactants for subsequent experiments. Nonylphenol ethoxylate NP9 has a nonylbenzene hydrophobic base and a long-chain polyoxyethylene hydrophilic layer. It can disrupt the aggregation of heavy crude oil components and reduce oil drop adhesion. It also has excellent hard water resistance and acid-alkali resistance, but its biodegradability is poor. The degradation product, nonylphenol, has moderate bioaccumulation and is toxic to microorganisms at high concentrations, potentially causing long-term environmental impacts. Polyoxyethylene acetylene glycol DY604, with its balanced hydrophilic-lipophilic structure, can significantly reduce oil-water interfacial tension and form stable emulsions. It has low toxicity and is easily biodegradable. However, its performance may be affected under high salinity conditions. Polyoxyethylene-polyoxymethylene formaldehyde SPO5 is synthesized by the condensation of polyoxyethylene alcohol and butyraldehyde. It has certain hydrophilic-lipophilic properties, can reduce oil-water interfacial tension, and works well with anionic surfactants. However, its biodegradability is average, and it may have adverse effects on the reservoir environment at high dosages. It may also undergo

chemical degradation under certain reservoir conditions, affecting its long-term stability. Cyclohexyl-2-ethylhexanoate TXIB contains polar groups and long-chain alkyls, making it amphiphilic. While reducing oil-water interfacial tension, it can also shorten the time required to establish miscible pressure. It has good biodegradability and low cost. However, its chemical stability may decrease under high temperatures or high pH conditions, and the ester group may undergo hydrolysis, affecting its performance. Additionally, its adsorption losses in the reservoir may lead to higher injection requirements in practical applications.

The microfluidic chip was fabricated using high-frequency hydrofluoric acid etching and thermal bonding techniques, with Borofloat33 borosilicate glass as the substrate, measuring $75 \times 75 \times 3.5$ mm. To ensure imaging clarity and pressure resistance, the chip consists of an etched layer (1.5 mm thick) and a cover layer (2.0 mm thick). The etched area on the glass plate ($75 \times 75 \times 1.5$ mm) was formed using wet etching with high-frequency hydrofluoric acid technology. Given the vulnerability of the chip's injection and outlet ends, a 2-mm-thick cover layer with two 2-mm-diameter holes (center-to-center distance of 90 mm) was used. After etching, the two glass plates were bonded via vacuum thermal compression to complete the chip. A layered reservoir substrate microfluidic chip model with more pronounced differences in pore size and connectivity was created. As shown in Figure 1. The etched area of the chip measures 12000×6000 μm , with an average pore size of 100 μm for the limestone layer and 30 μm for the shale layer, and an etching depth of 500 nm. The chip is primarily used to study the microscopic interaction mechanisms and displacement effects of CO_2 -surfactant composite systems with crude oil.

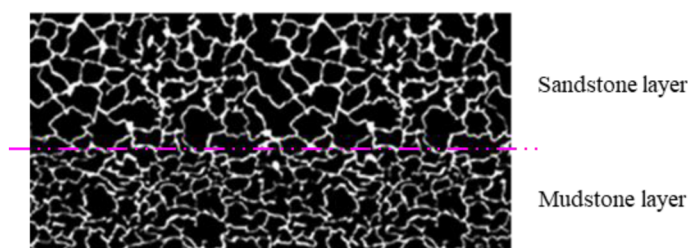


Figure 1—Microscopic visualization of etched glass chips.

Solubility Testing

The solubility of surfactants in CO_2 was tested using a fully automated interfacial tension instrument with a rotating drop method. As shown in Figure 2, a certain mass of surfactant was placed in a visual sapphire reactor and heated to 90°C . Once the temperature stabilized, sufficient CO_2 was injected into the intermediate container. The gas pressure was then increased to the required level using a gas pressurization system. The pressure changes in the reactor were monitored via a data acquisition system, and the mass changes of the surfactant at different pressures were recorded. The mass changes in the intermediate container were used to calculate the injected CO_2 mass, thereby determining the solubility of the surfactant in CO_2 at different pressures.

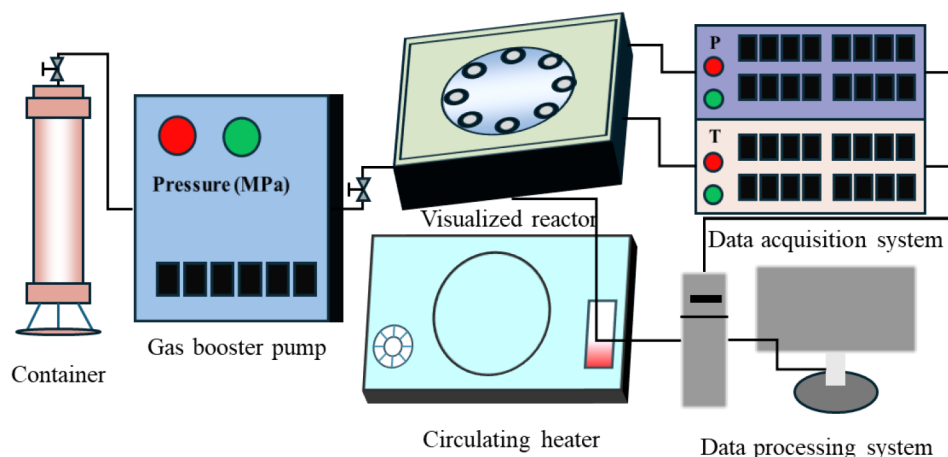


Figure 2—Surfactant solubility in CO₂ testing apparatus.

Interfacial Tension Experiments

The pendant drop method was used to measure interfacial tension. The basic principle involves determining interfacial tension using the Laplace equation based on digital drop contour data, which describes the pressure difference at the curved phase boundary as a function of curvature radius and interfacial tension.

In the interfacial tension measurement experiment, the interfacial tension instrument and sample cell were first cleaned with distilled water and ethanol to ensure a pollution-free testing environment. As shown in Figure 3, the crude oil was dehydrated and injected into the experimental medium, followed by pressurization to the experimental pressure. Surfactant solutions were prepared at a certain mass concentration. The surfactant solution was then injected into the measuring chamber, and crude oil was slowly injected at the probe to form a small oil drop for shape observation. The measuring tube was fixed on the instrument, and the temperature was set to 90°C. Using the pendant drop method, crude oil drops were injected into the surfactant water solution to form an oil-water interface. The camera system was activated, and once the liquid drop reached static equilibrium, the camera was adjusted to focus on the sample and a screenshot was taken. Finally, interfacial tension analysis software was used to record liquid drop shape changes and calculate interfacial tension.

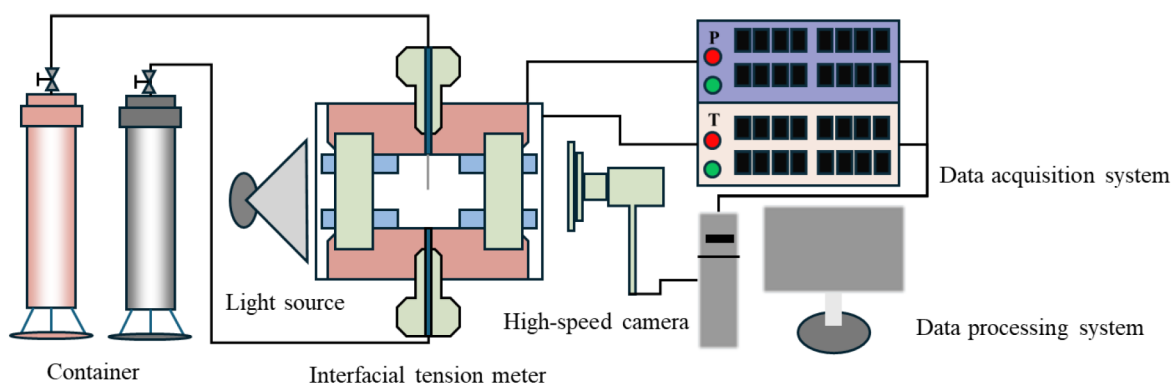


Figure 3—Visualized interfacial tension tester.

Miscible Pressure Experiments

Minimum miscibility pressure (MMP) is the lowest pressure at which a given reservoir oil can achieve miscibility with an injected gas of specified composition through a multi-stage contact process under reservoir temperature conditions. In this study, a microscopic visualized flow simulation experimental device (Figure 4) was used to measure the MMP of CO₂-surfactant composite systems with crude oil. The

principle involves using a microscopic visualized etched chip as a multi-contact channel for gas and oil. The environmental temperature was maintained at 90°C. After injecting crude oil into the etched chip, CO₂ and CO₂-surfactant composite systems were injected to test the MMP of different injection media with crude oil. Since temperature changes can affect MMP and the stability of the composite system, the temperature was kept constant during the process. Pressure was gradually increased from 5 MPa to the miscible pressure, and the pressure at this point was recorded as the MMP.

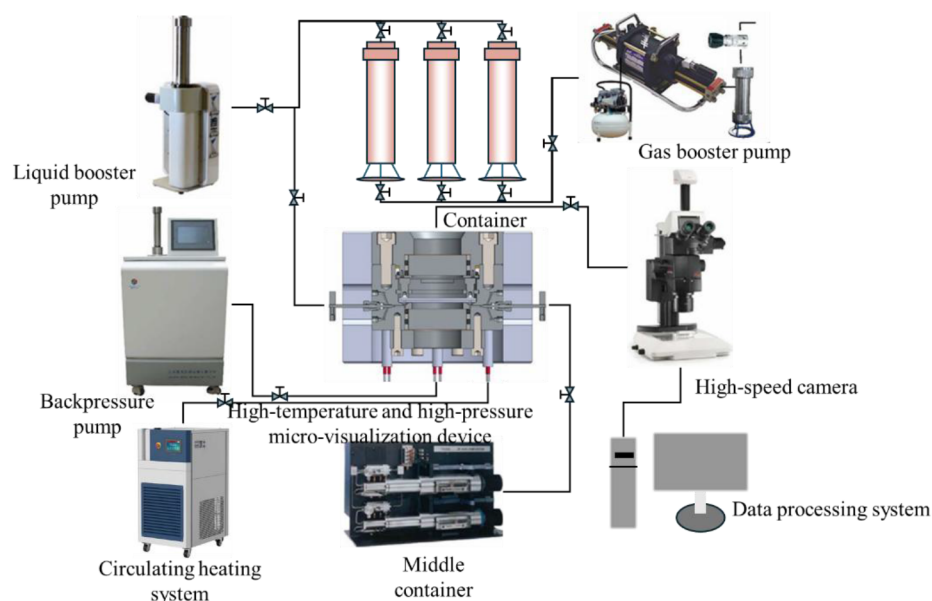


Figure 4—High-temperature and high-pressure micro-visualization displacement experimental device.

At the start of the experiment, the chip's confining pressure was subjected to vacuum treatment. After half an hour, the confining pressure water was saturated. The purpose of vacuuming the confining pressure was to prevent the formation of bubbles in the confining pressure after water saturation, which could affect the experimental results. The etched chip was then vacuum-treated, and the confining pressure was set to track the pressure difference mode, i.e., the confining pressure was set 2 MPa higher than the chip's inlet pressure. Two hours later, crude oil was injected into the intermediate container, followed by the injection of crude oil into the chip using a liquid pressurization pump. Once crude oil flowed out of the chip's outlet backpressure pump, the backpressure was increased to 15 MPa, and the inlet valve was closed, placing the chip in a sealed state. CO₂ was then injected into the intermediate container, and the pressure was increased to 15 MPa using a gas pressurization pump. The injection end was opened, equalizing the gas injection pressure with the chip's internal pressure. The gas injection rate was set to 0.002 mL·min⁻¹, and the fluid flow state in the visual glass window was continuously monitored from the data acquisition computer. Once gas was observed at the inlet, the backpressure was increased to 30 MPa to facilitate subsequent pressure changes for testing. The gas injection rate was then set to 0.004 mL·min⁻¹, and the changes in the gas-oil state within the chip were observed. When a color change was observed in the oil phase, the gas injection rate was reduced to 0.002 mL·min⁻¹, and the pressure increase rate was slowed to minimize experimental errors caused by human factors.

Composite System Displacement Experiments

CO₂ and CO₂-surfactant composite system micro-visual displacement experiments were conducted. The experimental device (Figure 4) was the same as that used for the MMP measurement. The oil saturation process prior to the experiment was identical to that of the MMP measurement. The experimental temperature was 90°C. The difference between the two methods was that after oil saturation in the chip

and oil emergence at the outlet, the backpressure was directly adjusted to 25 MPa, and the inlet valve was closed to seal the chip. The confining pressure was set to constant pressure mode to maintain a pressure of 27 MPa on the chip. The inlet vent valve was then opened, and toluene was injected into the pipeline between the gas injection intermediate container and the chip inlet valve to clean it. Residual toluene in the pipeline was removed with gas. Once the experimental device's temperature stabilized, gas injection began. When the gas pressure reached 25 MPa, the gas injection rate was set to $0.002 \text{ mL} \cdot \text{min}^{-1}$, and the gas-oil flow state in the chip was observed using a high-speed camera. The experiment concluded when the crude oil in the chip ceased to flow.

Results and Discussion

Surfactant Solubility in CO_2

Throughout the experiment, other conditions were kept consistent, with only the pressure varied. As shown in Figure 5, the solubility of the four surfactants in CO_2 increased with pressure, though the growth trends differed. According to Henry's Law, surfactant solubility in CO_2 increases as the partial pressure of CO_2 rises in a closed system. Initially, CO_2 exists as a gas, so the solubility increase with pressure is relatively slow. Once CO_2 enters the supercritical state, surfactant solubility in CO_2 rises significantly. In the supercritical state, DY604, SPO5, and NP9 showed marked increases in solubility with pressure. As pressure increases in the composite system, CO_2 and surfactant molecules are forced closer, enhancing intermolecular attraction. As shown in Figure 6 CO_2 molecules surround surfactant molecules more tightly, forming a solvation effect that aids surfactant dispersion in CO_2 . The sudden acceleration of solubility with pressure in the supercritical state is due to the local density of CO_2 around solute molecules exceeding the bulk density, particularly near the critical pressure. This faster increase in solvent density around solute molecules enhances solvation capacity, causing solubility to rise sharply near the critical pressure. DY604 and NP-9 have non-polar methyl groups in their tail chains with relatively high surface energy. CO_2 , also non-polar, dissolves more readily in these non-polar regions according to the like-dissolves-like principle. This structural and property match allows CO_2 molecules to contact and bind more closely with surfactant tail chains, increasing CO_2 solubility in these surfactants.

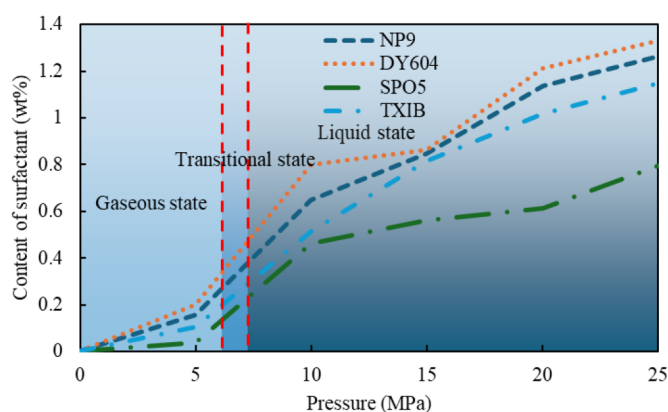


Figure 5—The solubility of surfactants in CO_2 under different pressure conditions.

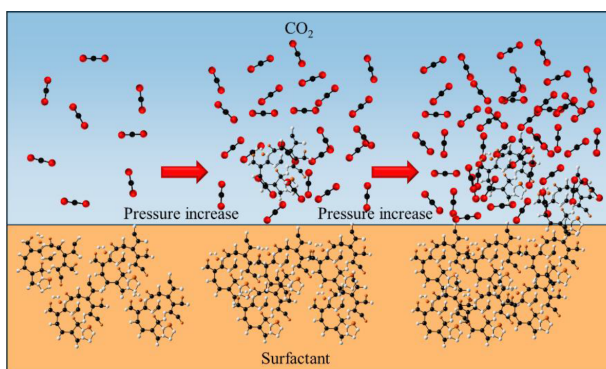


Figure 6—Schematic diagram of the solubility behavior of surfactants in CO₂ with increasing pressure.

CO₂ and Composite Systems Interfacial Tension with Crude Oil

The interfacial tension between five injection media and crude oil at 90°C was measured, with results shown in Figure 7. As pressure increased, gas-oil interfacial tension gradually decreased. This was mainly because higher pressure 促使 CO₂ to dissolve further into crude oil, blurring the interface between them. With increasing pressure, CO₂ molecules penetrated deeper into crude oil, interacting with its molecules and weakening the intermolecular forces on the crude oil droplet surface, thereby reducing interfacial tension. From the figure, the gas-oil interfacial tension in pure CO₂ experiments was higher than that in CO₂-surfactant composite systems. This may be related to surfactant adsorption and arrangement at the interface. Surfactant molecules, with their unique amphiphilic nature, spontaneously aggregate at the gas-liquid interface to form ordered molecular layers. This behavior alters the molecular arrangement and energy state at the interface. The hydrophilic and lipophilic groups of surfactants extend toward the water and oil phases, respectively, forming an ordered adsorption layer. This disrupts the original intermolecular forces at the oil-gas interface, reduces the mutual attraction between oil and gas, and thus lowers interfacial tension. Comparing the interfacial tension of the four composite systems with crude oil, CO₂-NP9 composite systems showed the greatest reduction in interfacial tension at relatively low pressures. This may be due to NP9 surfactant's long alkyl chain and benzene ring, which dominate adsorption at the oil-gas interface. After adsorption, the lipophilic groups interact more strongly with crude oil molecules, and the hydrophilic groups interact more strongly with CO₂ molecules. This helps form a more stable interfacial adsorption layer and reduces interfacial tension. As pressure increased, DY604 showed better adaptability under high pressure, with CO₂-DY604 composite systems achieving the greatest reduction in interfacial tension. Both CO₂-DY604 and CO₂-NP9 composite systems demonstrated good interfacial tension reduction effects at high pressures.

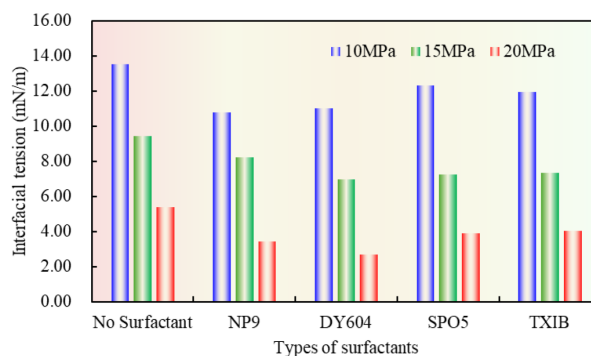


Figure 7—The interfacial tension between CO₂ and crude oil, as well as between the CO₂-surfactant complex system and crude oil, under different pressure conditions.

Minimum Miscibility Pressure Microfluidic Experimental Results

Compared to conventional MMP measurement devices, the method using a microscopic visualized experimental device allows direct observation of the miscible interface. Based on the observed changes in crude oil state and gas-oil flow patterns, the variations in crude oil components during gas-oil miscibility can be analyzed. Additionally, this method can simulate the temperature and pressure conditions of actual reservoirs, providing a more accurate simulation of gas-oil MMP in underground reservoirs. As shown in Figure 8, the MMP of pure CO₂ with sample crude oil was 27.98 MPa, with the CO₂-DY604 composite system showing the best MMP reduction effect.

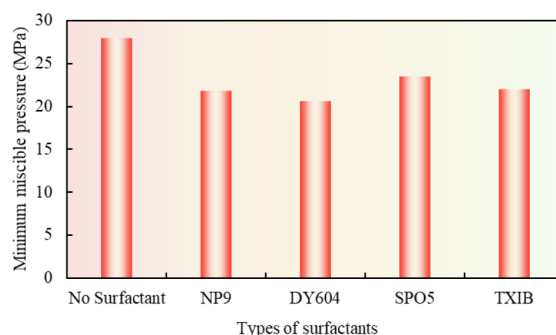


Figure 8—The minimum miscibility pressure (MMP) of CO₂ and the CO₂-surfactant complex system with crude oil under different pressure conditions

The experimental temperature was 90°C, consistent with previous experiments, essentially ruling out temperature effects on the results. The MMP was 20.62 MPa, close to the 20 MPa in the interfacial tension experiment. However, no gas-oil miscibility was observed at this pressure in the interfacial tension experiment, possibly due to the 0.62 MPa pressure difference or fluid heterogeneity. The compositions of crude oil and CO₂ may not be uniform, affecting their interaction and mixing process. Differences in the volatility and solubility of crude oil components can lead to local mixing inhomogeneity. Even near MMP, complete mixing may require longer times or higher pressures. More likely, pore size significantly impacts gas-oil interactions. In microscopic visualized experiments, the gas-oil flow channels resemble those in actual reservoirs. Smaller pores increase specific surface area per unit volume, enhancing contact between the composite system and pore walls. The adsorption of composite fluids onto pore walls forms a layer that alters the composite system's properties, reduces gas-oil interfacial tension, and enables easier gas dissolution into liquids, thereby lowering MMP.

Micro-Visualized Displacement Experimental Results

In CO₂ and CO₂-surfactant composite system micro-displacement experiments, the effects of the composite system on crude oil components during contact were analyzed (Zhang and Han 2013). Micro-displacement efficiency for five injection media was determined based on residual oil distribution areas post-displacement. As shown in Figure 9(a), during contact between the composite system and crude oil, the oil phase volume expanded, with crude oil gradually swelling toward the gas phase. Near the gas-oil contact zone, crude oil closer to the gas phase appeared lighter in color, while that near the oil phase underwent color changes from light to dark and then back to light over time. At t=12 s, crude oil in channels distant from the gas-oil mixing interface appeared darker than at t=0 s, indicating light component extraction by the gas phase, increasing heavy component proportions and deepening oil phase color. From t=12 s to t=24 s, the oil phase near the gas-oil contact zone gradually lightened as the composite system diffused, reducing heavy component proportions (Zhang, Guo, et al. 2024). In contrast, crude oil distant from the mixing interface saw heavy components adhere to pore walls, with their deposition in the oil phase lowering heavy component proportions, resulting in color lightening. At t=36 s, the oil phase near the gas-oil contact zone appeared

darker, possibly due to longer contact with the gas phase and stronger fluid flow capacity compared to distant channels. However, the greater density of heavy components introduced flow resistance. Overall, heavy component accumulation occurred in the planar flow channels during gas-oil contact, causing deeper coloration in these regions.

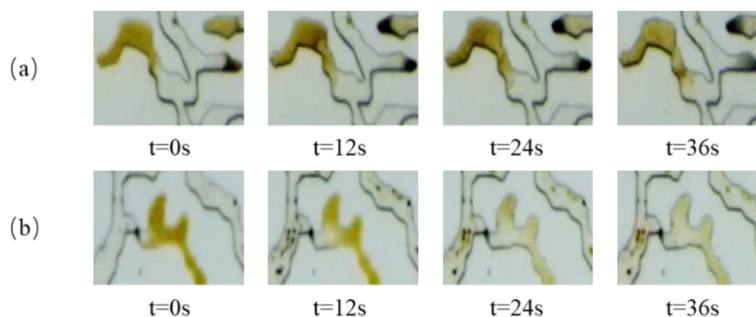


Figure 9—Two representative diagrams of the mechanisms of oil-gas two-phase interaction during the displacement stage.

Figure 9(a) primarily illustrates the variation patterns of gas-oil two-phase components in non-displaceable regions, while Figure 9(b) showcases the changes in gas-oil two-phase components during miscible sweep in displaceable regions. At $t=36$ s, compared to $t=0$ s, the left channel exhibits the precipitation of heavy components from the oil phase, while the right channel shows increased miscibility in the gas-oil contact zone (Hermansyah et al. 2024). From $t=0$ s to $t=36$ s, the oil phase color gradually lightens to a pale yellow or off-white. During the sweep process, light components in the oil phase dissolve into the gas phase, and gas components diffuse into the oil phase. This contact effectively reduces crude oil viscosity and mitigates the impact of viscous forces on oil flow.

As the gas phase flows, the miscible front continues to advance, gradually sweeping corner crude oil, with the sweep extent reaching maximum as flow progresses. Furthermore, the effects of pure CO_2 , CO_2 -NP9, CO_2 -DY604, CO_2 -SPO5, and CO_2 -TXIB on crude oil recovery were analyzed. Figure 10 shows that pure CO_2 achieved only 51.2% recovery, while CO_2 -surfactant composite systems showed varying degrees of improvement. CO_2 can mix with crude oil, reducing viscosity and enhancing flowability. Surfactants lower interfacial tension required for gas-oil miscibility, synergistically improving displacement efficiency. Regarding displacement efficiency, CO_2 -DY604 composite systems performed best, followed by CO_2 -NP9, CO_2 -TXIB, and CO_2 -SPO5 (Dong et al. 2024). The high dielectric constant of high-pressure CO_2 ($\epsilon > 1.5$) shields charge repulsion, compresses solvation shells to reduce molecular spacing, and forms a dense monolayer, significantly lowering interfacial tension (Gong et al. 2024). DY604's molecular structure likely excels in reducing gas-oil interfacial tension. Its balanced lipophilic and hydrophilic groups enable effective adsorption at the oil-gas interface, forming a stable layer that significantly reduces interfacial tension and enhances displacement efficiency (Kuang et al. 2021). NP9's lipophilic group length or shape may make its interface adsorption less efficient than DY604, resulting in relatively lower displacement efficiency. TXIB's molecular structure may limit its adsorption capacity at the oil-gas interface (Wang et al. 2023). Its balance of lipophilicity and hydrophilicity may be inferior to DY604 and NP9, leading to suboptimal interfacial tension reduction (Lee et al. 2021). Although SPO5 shows some interfacial tension reduction, its shorter lipophilic group may fail to establish strong interactions with crude oil molecules. Additionally, SPO5's lower solubility may hinder its sufficient reach to the interface during displacement, affecting its interfacial tension reduction capability. In summary, DY604's molecular structure, with its balanced lipophilic and hydrophilic groups and superior interfacial adsorption capacity, outperforms other surfactants in displacement efficiency. NP9, TXIB, and SPO5, due to structural differences, exhibit progressively weaker lipophilicity, hydrophilicity, and interfacial adsorption capacity, leading to gradually decreasing displacement efficiency.

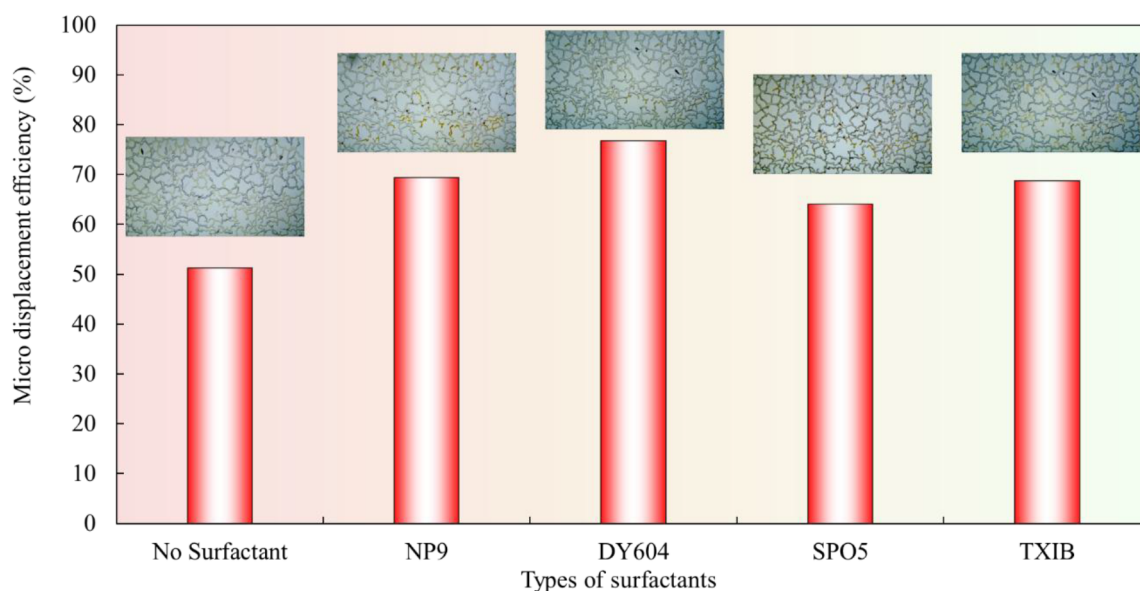


Figure 10—The microscopic oil displacement efficiency corresponding to different injection media.

Summary and Conclusions

The CO₂ plus surfactant technology demonstrates multifaceted application potential in the field of CCUS (Carbon Capture, Utilization, and Storage), enhancing carbon utilization efficiency, oil displacement effects, and resource transformation value. Constructing a CO₂-surfactant complex system for oil extraction not only improves CO₂ utilization rates but also increases oil recovery rates. The main research conclusions regarding the oil displacement aspect of the CO₂-surfactant composite system are as follows:

Under consistent experimental conditions with varying pressure, the solubility of four surfactants in CO₂ increased with pressure. According to Henry's Law, higher pressure in a closed system increases CO₂'s partial pressure, thereby raising surfactant solubility in CO₂. Once CO₂ reaches the supercritical state, the solubility growth trend becomes more pronounced. DY604 and NP-9, with non-polar methyl groups in their tail chains and relatively high surface energy, dissolve more readily in non-polar CO₂ molecules based on the like-dissolves-like principle. Thus, these two surfactants exhibit higher solubility in CO₂ than the others.

Interfacial tension measurements between five injection media and crude oil at 90°C revealed a gradual decrease in gas-oil interfacial tension with increasing pressure. This is attributed to CO₂ dissolving further into crude oil, weakening intermolecular forces on the crude oil droplet surface. Surfactant molecules, with their amphiphilic nature, adsorb at the gas-liquid interface to form ordered molecular layers. This alters the interface's molecular arrangement and energy state, disrupts original intermolecular forces at the oil-gas interface, and reduces interfacial tension.

Compared to conventional MMP measurement devices, the method using a microscopic visualized experimental setup allows direct observation of the miscible interface. By analyzing changes in crude oil state and gas-oil flow patterns, variations in crude oil components during gas-oil mixing can be assessed. Additionally, this method can simulate actual reservoir temperature and pressure conditions, providing a more accurate representation of gas-oil MMP in underground reservoirs. Experimental results indicated that the MMP between pure CO₂ and sample crude oil was 27.98 MPa, with the CO₂-DY604 composite system showing the best MMP reduction effect. Furthermore, experiments revealed that pore size impacts gas-oil interactions. In microscopic visualized experiments, gas-oil flow channels resemble those in actual reservoirs. Smaller pores increase specific surface area per unit volume, enhancing contact between the composite system and pore walls. This strengthens the adsorption of composite fluids onto pore walls,

forming an adsorption layer that affects the composite system's properties. The layer reduces gas-oil interfacial tension, enabling easier gas dissolution into liquids and thus lowering MMP.

In CO₂ and CO₂-surfactant composite system micro-displacement experiments, the oil phase volume expanded upon contact with the composite system, with crude oil gradually swelling toward the gas phase. Changes in crude oil color were observed in the gas-oil contact zone. Light components within crude oil were extracted by the gas phase, increasing the proportion of heavy components and deepening the oil phase color. Subsequently, the oil phase color in the gas-oil contact zone gradually lightened as the composite system diffused, reducing the relative content of heavy components. The study also analyzed the effects of different injection media on crude oil recovery. Pure CO₂ achieved a recovery rate of only 51.2%, while CO₂-surfactant composite systems showed varying degrees of improvement. CO₂ can mix with crude oil, reducing viscosity and enhancing flowability. Surfactants lower the interfacial tension required for gas-oil miscibility, synergistically improving displacement efficiency. Among the composite systems, CO₂-DY604 demonstrated the highest displacement efficiency, followed by CO₂-NP9, CO₂-TXIB, and CO₂-SPO5. This is mainly because DY604's molecular structure excels in reducing gas-oil interfacial tension. Its balanced lipophilic and hydrophilic groups enable effective adsorption at the oil-gas interface, forming a stable layer that significantly reduces interfacial tension and enhances displacement efficiency. NP9, TXIB, and SPO5, due to structural differences, exhibit progressively weaker lipophilicity, hydrophilicity, and interfacial adsorption capacity, leading to gradually decreasing displacement efficiency.

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